

$S^n, \cancel{S^1}, \mathbb{R}P^n, \text{---}$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

$M \cup N$ $\wedge \vee$ ---
 π_0, π_1^2

$\mathbb{Z} \times \mathbb{Z}$ klein. free product.
 $\langle a, b \mid aba = b \rangle$ $ababab...$

$aba^{-1}b^{-1} = e$ ---
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^2$ ---
 $\pi_1 = \mathbb{Z} \times \mathbb{Z}$ $\pi_2 = \mathbb{Z} \oplus \mathbb{Z}$

Theorem $M: n\text{-dim}$ ---
 $\text{with } (n \geq 2)$

$\pi_1(M) \cong \pi_1(M \cup \bullet)$ ---
 $\pi_1(\mathbb{R}P^n) = \mathbb{Z} \quad (n \geq 2)$

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